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Original Article

Direct healthcare costs in the first 2 years of life: A comparison of screened and clinically diagnosed children with cystic fibrosis – The Irish comparative outcomes study of CF (ICOS)

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A R T I C L E I N F O

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A B S T R A C T

Background: In July 2011, Cystic Fibrosis (CF) was added to the Newborn Bloodspot Screening Programme in Ireland. The Irish Comparative Outcomes Study (ICOS) is a historical cohort study established to compare outcomes between clinically-detected and screen-detected children with CF. Here we present the results of economic analysis comparing direct healthcare costs in the first 2 years of life of children born between mid-2008 and mid-2016, in the pre-CF transmembrane conductance regulator modulator era.

Methods: Healthcare resource use information was obtained from Cystic Fibrosis Registry of Ireland (CFRI), medical records and parental questionnaire. Hospital admissions, emergency department visits, outpatient appointments, antibiotics and maintenance medications were included. Costs were estimated using the Health Service Executive Casemix, Irish Medicines Formulary and hospital pharmacy data, adjusted for inflation using Consumer Price Index data from the Central Statistics Office. A Negative Binomial regression was used, with time in the study as an offset.

Results: Overall participation was 93 %. After exclusion of those with meconium ileus, data from 139 patients, with follow-up to 2 years of age, were available. 72 (51.8 %) were from the clinically diagnosed cohort. In the final model (n=105), clinically diagnosed children had 2.62-fold higher costs per annum ($p<0.0001$), when adjusted for confounders, including homozygous ΔF508 or G511D mutation, socio-demographic factors and time between diagnosis and first CFRI interaction.

Conclusions: There are few studies evaluating economic aspects of newborn screening for CF using routine care data. These results imply that the benefits of newborn screening extend to direct healthcare costs borne by the State.

1. Introduction

Cystic fibrosis (CF) is the most common fatal genetic disease in populations of Northern European ancestry [1] and Ireland has the highest incidence in the world (approximately 1/1353) [2]. It is a severe multisystem condition which is associated with significant costs to the

Abbreviations: ICOS, Irish comparative outcomes study of CF; CFRI, cystic fibrosis registry of Ireland; NBS, newborn screening; IRT, immunoreactive trypsin; HSE, health service executive; HPO, healthcare pricing office; BAL, bronchoalveolar lavage; TOBI, tobramycin for inhalation.

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healthcare system including costs of routine assessments, hospital admissions and visits, emergency department (ED) visits, medications etc. [2,3] Given that the health effects of this condition start early in life and early treatment can be initiated [4], newborn screening (NBS) has been adopted in many countries worldwide [3] and was introduced in Ireland in 2011. A 2009 Cochrane systematic review of NBS for CF found growth and nutritional benefits in the screened cohort although the long-term pulmonary benefits remained unclear. It also found potential cost savings of achieving diagnosis by screening as compared with clinical diagnoses [4].

The Irish Comparative Outcomes Study (ICOS) was established in 2013 to examine the impact of the introduction of CF NBS in Ireland. It is a national historical cohort study which aims to compare outcomes between clinically- and screen-diagnosed children with CF. The screening programme for CF measures the plasma level of immunoreactive trypsin (IRT); IRT alone has sensitivity 83.33 %; specificity 99.77 %. However, inclusion of DNA analysis improves sensitivity to 96.15 %, with specificity of 99.75 % [5].

The study has previously found improved growth and longer time to *Pseudomonas aeruginosa* acquisition in screened children [6]. While clinical outcomes are clearly important, the economic implications of screening are less frequently examined using routine care data. A number of international studies have conducted economic evaluation of newborn screening for CF and have reported lower total cost, hospitalisation and treatment costs [4,7]. In Ireland two studies have evaluated costs, however these are now out of date, and they did not look at direct healthcare costs or address the comparison between clinically diagnosed and screen detected children [2,8]. In this study, we sought to conduct an economic analysis examining direct healthcare costs.

2. Aims

The aim of this study was to estimate the direct healthcare costs of children with CF in the first 2 years of life and to compare these costs between those diagnosed clinically and those diagnosed via newborn screening.

3. Material and methods

3.1. Recruitment of study participants

The Irish Comparative Outcomes Study is a national historical cohort study. The two cohorts that constitute this study were delineated by date of birth, with the clinical cohort born in the three years prior to the introduction of screening (1st July 2008 to 30th June 2011) and the screened cohort born in the five years after the introduction of screening (1st July 2011 to 30th June 2016). Inclusion of clinically diagnosed children, who act as the historical control group, was limited to only the recent past in order to avoid chronology bias. Children were recruited from each of the six specialist centres for CF in Ireland (specialist centres for CF in Ireland include: Our Lady's Children's Hospital, Crumlin; The National Children's Hospital, Tallaght; The Children's University Hospital, Temple Street; University Hospital Galway; University Hospital Limerick; Cork University Hospital) after ethical approval was obtained from each centre. Informed consent was obtained from parents and a total of 232 children with CF were enrolled in the study.

3.2. Collection of cost data

A bottom-up micro-costing approach was taken. Data on healthcare resources utilisation were available from 3 sources, both retrospective and prospective. 1) Parents of the enrolled children with CF were asked to complete an economic questionnaire (Appendix A and B) as their consents were received, which asked about healthcare resources used and costs incurred prior to diagnosis and collected demographic information. 2) Post diagnosis data on healthcare resource utilisation were provided on registered patients by the Cystic Fibrosis Registry of Ireland (CFRI). 3) Data were collected from hospital charts by ICOS researchers. Using these three sources, we were able to ensure that relevant pre and post diagnosis costs were included, so as not to under- or over-cost either cohort.

The direct healthcare costs included in the analysis were: 1) Costs that occur pre and post diagnosis: inpatient hospital admissions, public outpatient appointments and acute courses of oral antibiotics. 2) Costs that occur post diagnosis: day-case hospital admissions, acute courses of intravenous (IV) antibiotics, inhaled antibiotics (for eradication of pseudomonas aeruginosa) and regular maintenance medications. 3) Pre-diagnosis only costs: ED visits. Post diagnosis ED visits were not available and not included, as children with CF are generally managed by specialist teams once diagnosed, avoiding ED visits to a large extent.

3.3. Adjusting for inflation

2011 was chosen as the base year for assignment of costs. Costs were then adjusted for inflation using Consumer Price Index data from the Central Statistics Office [9]. Where an exact date for the cost incurred was available, this year was used for adjustment. Where one was not available, an estimate was made by applying the year of birth of the child for adjustment purposes.

3.4. Perspective of the economic analysis

Ireland has a two-tier health care system, with both private and public elements. The administrator of public healthcare is the Health Service Executive (HSE). The perspective taken in this analysis was that of costs incurred by the public healthcare system and thus, the State. Prior to diagnosis, some patients were liable for costs of visits to private specialists, as well as medications, while others had these paid for by the HSE. Parents who held medical cards (35.5 % of the population in 2016) [10] would not have had to pay any fees to hospitals or General Practitioners (GPs) and there would be no fee for prescriptions; these costs would have been borne by the HSE. Those who did not hold a medical card would have had to pay a total of €200 per annum hospital costs as well as GP fees and the cost of prescription medications (subject to a fixed maximum per family). Once diagnosed with CF, these costs were incurred by the HSE under the Long-Term Illness Scheme [11]. Thus, where patients paid for the costs of appointments or medications pre-diagnosis, these were not included as costs to the public healthcare system. Hospital admissions were all included and costed as public.

3.5. Attribution of costs to the collected data

Inpatient hospital admissions, day-case hospital admissions, ED visits and public outpatient appointments were costed according to national average data obtained from the HSE [12] and personal correspondence with the national Healthcare Pricing Office (HPO). These were: €825 per inpatient day (HPO) €735 per day-case with bronchoscopy, €637 per day case without bronchoscopy, €268 per ED visit and €130 per outpatient visit (HSE).

In terms of physician costs, all children with CF get free physician care through the HSE public healthcare system. Physician costs are included in the estimation of the cost of inpatient stays, outpatient appointments, day cases and emergency department visits. For all public OPD appointments, children would have been seen by a consultant paediatrician; in the pre-diagnosis OPD that might be a general paediatrician or respiratory paediatrician, in post-diagnosis OPD it would be a consultant respiratory physician specialising in the care of children with CF. However, the cost applied was the same regardless.

Costs of medications were obtained from the Irish Medicines Formulary or a hospital pharmacy [13]. Where a price range was given, the midrange cost was used. Where no price was available for the oral suspension, the price per tablet of the dose closest was chosen. For IV antibiotics, the cost per dose was chosen as the vial size closest to the mean or modal dose from the data. Where drug doses and length of administration were unclear, decisions were taken based on advice given by a specialist CF physician, and on consulting drug formularies or specialist CF treatment protocols.

Where a course of oral antibiotics (unspecified type) was reported in the parental questionnaire, the cost of one week of amoxicillin 125 mg three times a day was applied. For oral antibiotic data collected from the hospital charts, the cost for the typical dose and frequency for a one-year-old child was applied. For inhaled antibiotics, the cost of 28 days of nebulised tobramycin treatment for each positive pseudomonas aeruginosa culture seen at least 28 days apart was applied. Infants and young children with CF are screened frequently (at three monthly CF clinic visits, at a minimum), typically using cough swabs, with annual bronchoalveolar lavage (BAL) used in some centres, looking for airway pathogens that require treatment. If *Pseudomonas aeruginosa* is detected it is always treated, with nebulised antibiotics, typically using tobramycin for inhalation (TOBI) for 28 days as part of an eradication regimen. The presence or absence of bronchiectasis does not influence the decision to treat.

For maintenance medications, only maintenance oral antibiotics (presumed to be flucloxacillin) and pancreatic enzyme replacement therapy were included and were costed at the usual dose and frequency for a one-year-old child. Where a patient was documented as ever being on one of these therapies, the cost of continuous therapy from date of diagnosis to the age of 2 was applied.

4. Statistical methods

Descriptive statistics for each group were calculated using within-group percentages, and medians for ages and durations. Cost components were categorised into zero, between €1 and €1000, and over €1000 as convenient ranges for understanding the distribution between the NBS and Clinically diagnosed cohorts. Pearson chi-square test and Mann-Whitney test p-values are presented. A Negative Binomial generalised linear model was used to model the skewed, over-dispersed (assessed by the model deviance) cost variable (using cost as a continuous variable) using a natural log link function and time in the study as an offset. Robust (sandwich) standard errors were used. Where data on siblings were collected only the index case was used. The models were fit using clinical versus NBS diagnostic method as the primary predictor, and adjusted models included the mean-centred duration between diagnosis and the first encounter with the CFRI, and indicator variables were the gender of the child, whether any parent smoked, whether the child was homozygous for either del-F508 or G511D mutations, and whether the mother held a third level (or higher) degree. We also investigated the performance of a range of models including Poisson, Negative Binomial, linear regression with a log-transformed costs variable, linear regression with lognormal errors, and Gamma regression. We noted overdispersion affecting the Poisson model, and high sensitivity of the log-transformed and lognormal models to the value of an increment to allow for zero costs (ranging from +€0.5 to half the size of the lowest non-zero cost = +€600) in the direction and the magnitude of the difference between the Clinical and NBS groups. A Gamma regression model yielded near-identical numerical results and conclusion to the planned Negative Binomial regression. Sensitivity analyses were conducted to investigate subgroups or influences of interest including clinically diagnosed limiting to those diagnosed after 2 months of age and clinically diagnosed limiting to those without a known sibling with CF. The analysis was conducted using R statistical software version 4.04 [14], with packages MASS [15], Zelig [16], lmtest [17], sandwich [18], and logNormReg [19].

5. Results

5.1. Study population

A total of 232 patients were recruited into the ICOS study (93 % participation rate). As per previous ICOS analyses, patients diagnosed outside Ireland (n= 4), those with a diagnosis of CF screen positive inconclusive disease (n=2) and those with a history of meconium ileus (n=40) were excluded, as in prior studies [20]. After further exclusion of those whose last date of follow-up with the CFRI was before the age of 2 or who were not registered with the CFRI (n=47), 139 patients were included in the analysis. 67 (48.2 %) were from the NBS cohort and 72 (51.8 %) from the clinically diagnosed cohort. Descriptive information for our sample is presented in Table 2.

Those in the clinically diagnosed cohort were on average older at diagnosis and less likely to have been diagnosed before the age of 2. There were no significant differences in gender, prevalence of ΔF508 homozygous genotype, or age at which first and last data were available from the CFRI between the groups.

5.2. Initial comparison of costs

The costs between the groups were then compared, broken down into the following categories: €0, €1 - €1000 and >€1000. The results are displayed in Table 3. As can be seen, there were significant differences between the groups in all categories except for IV antibiotics and inhaled antibiotics. The clinical group had significantly greater costs in inpatient admissions and pre-diagnosis ED visits, while the screened group had significantly higher costs in the following categories: day cases (procedures), public outpatient appointments, acute oral antibiotics and maintenance medications. The mean costs per group are shown in Table 4.

As shown in Table 4, the biggest difference in absolute costs is seen in the cost of inpatient admissions between the two cohorts, with the clinical group having a mean cost of €10,940 as compared with €5070 in the screened group.

5.3. Regression models

The results of the unadjusted and adjusted regression models are shown in Table 5.

The multivariable analysis presents the ratio of the total costs for the clinical cohort as compared with the NBS cohort per unit time in study. Clinically diagnosed children had 2.62-fold higher costs per annum than those diagnosed by NBS, when adjusted for a range of potential confounders and factors likely to influence disease severity.

A range of sensitivity analyses were conducted to investigate subgroups or influences of interest on these findings. Firstly, when the clinically diagnosed group were limited to those diagnosed after 2 months of age (n=45), the findings are somewhat attenuated, with an unadjusted IRR of 3.10 (95 % CI: 1.99, 4.82; $p<0.0001$), and an adjusted IRR of 2.44 (95 % CI: 1.55, 3.86; $p=0.0001$).

Among the clinically diagnosed children, 11 had a known sibling with CF diagnosis. Limiting the clinically diagnosed children to those without a known sibling with a CF diagnosis (n=128), as a proxy for family history, increased the group difference relative to NBS-diagnosed children, with an unadjusted IRR of 4.44 (95 % CI: 2.74, 7.21; $p<0.0001$), and an adjusted IRR of 3.15 (95 % CI: 1.95, 5.08; $p<0.0001$).

6. Discussion

Ireland is a small country with a high incidence of CF. The Irish healthcare system is thus disproportionately affected by the costs of this severe multisystem disease, and any clinical or cost differential between screened and clinically diagnosed children is important to demonstrate in the Irish context. This study is unique in the CF cost literature in that it was a national comparative cohort study with almost complete coverage, and three data sources were used to ensure completeness of the data. Many studies in the literature are unable to do this and include registry data only [8,21] or retrospective survey data only [22,23]. One of the strengths of this study is the representativeness of the study sample. This is a national study with a very high response rate, and we included parents of children from all the CF-specialist centres in Ireland; hence reflecting data from both rural and urban participants. Although this was a census study of children with CF in Ireland with a participation rate of almost 94 %, we only had sufficient numbers of children in the 0–2 years to make this comparison. There were smaller numbers of children aged 3–5. We also note that when aggregating costs for families, we had missing data - predominantly in the clinically diagnosed cohort. While no differences in mutation, or socio-environmental factors were seen between those missing costs, and not, this remains a potential source of unresolvable bias in the findings.

This study also provides an important comparison between NBS and clinically diagnosed children. The results show that clinically diagnosed children incurred 2.62 times more costs than NBS children on public healthcare, after adjusting for covariates. The results also show that clinically diagnosed children remained more costly when limiting the analysis to those diagnosed after 2 months of age and to those without a known sibling with a diagnosis of CF. These are the children who have the most opportunity to benefit from screening, as they would not have been identified early due to severe disease or a family history [3].

The biggest cost differential comes from the much higher cost attributed to hospital admissions in the clinical cohort. This may be due to increased number of admissions for clinical symptoms prior to being diagnosed with CF, or increased disease severity after diagnosis. The higher costs of day cases, public outpatient appointments, acute oral antibiotics and maintenance medications in the NBS group may reflect their earlier diagnosis and subsequent commencement on standard monitoring and treatment. It is noteworthy that these costs are overshadowed by the savings from reduced hospital admissions.

In fact, the results likely underestimate the difference in costs between screened and clinically diagnosed, as nearly a third of the clinically diagnosed cohort had not been diagnosed before the age of 2 and thus would not have incurred routine hospital costs including outpatient follow-up, day cases and maintenance medications.

This analysis did not set out to include indirect costs to families of children with CF, but it is likely that these are also reduced in the screened group, especially given the lower number of hospitalisations [3]. It is known that direct medical costs account for 50–70 % of total medical costs based on CF costing studies, with non direct medical and indirect nonmedical (e.g. sick days, lost work) contributing a significant component [24,25]. As such, one can hypothesise that NBS will likely also reduce these other costs and be of economic benefit on both a patient and societal level (in addition to benefits in clinical outcomes).

We did not include post diagnosis ED visits. While we recognise that ED visits can be an important source of healthcare utilisation, the number of visits to ED for children with CF attending specialist CF centres in Ireland is kept to an absolute minimum. The risk of cross-infection is considered high in the ED environment and all efforts are made to minimise the need for ED attendance. This is achieved largely by way of an "open door" policy in each CF centre whereby parents have direct phone access to the CF unit 9am to 5pm, five days a week, with rapid same day in person reviews possible when indicated. Although not a 24 $\times$ 7 service, this approach appears to be largely successful with most children with CF never having attended the ED.

These findings are consistent with those from other countries such as the UK, where Sims et al. demonstrated that clinically diagnosed patients with CF up to 9 years of age received treatment costing approximately 60–400 % more than screened children [3].

The results shown here reflect the first part of a national historical cohort study which is continuing. Part 2 is ongoing and follows children with CF to age 8–11. Follow up to age 8–11 will also allow greater comparison of respiratory costs as older children can undertake spirometry and other investigations. It will be of interest to extend the cost analysis to see if the benefits of screening demonstrated here are borne out in the longer term, as well as considering other issues such as quality of life. Given that screening was only introduced in Ireland in 2011, it will take time to see the effect on survival.

It is important to consider recent developments in CF care which may alter the impact of screening. In the Irish context, CFTR modulator drugs have recently become available. Children from age 2 with G551D mutation can be prescribed Ivacaftor and those homozygous for ΔF508 mutation can be prescribed lumacaftor/Ivacaftor from age 6; Kaftrio (Lumacaftor/Ivacaftor/Exacaftor) has recently been introduced for homozygous or heterozygous ΔF508 mutation. These drugs are extremely expensive but will likely also have beneficial effects on costs in terms of reduced exacerbations/hospitalisations as well as other important factors such as improved quality of life. The current analysis was conducted in the pre-CFTR modulator era and serves as a baseline against which analyses that consider the increased costs of these drugs in screened children can be compared.

This study had some limitations. Firstly, not all possible costs were included. Of those included, some variables were only available from the CFRI and not supplemented by ICOS hospital chart data. Furthermore not all children had data from all three sources used, although this was minimal. Registry data is known to be limited in that data collection starts at registration (usually diagnosis), thus data prior may not be captured. We attempted to account for this in our analyses by supplementing with chart review; clinical data collection for pre-diagnosis management from General Practitioners and Public Health Nurses and parental questionnaires and including a variable which measured the time from diagnosis to first registering with the CFRI. Furthermore, estimates on out-of-pocket expenses are solely based on parents' reports so recall bias cannot be excluded. While some parts of the study included estimates of utilisation data, we went to quite extensive lengths to calculate exact costs per child. Data were obtained from three sources: hospital charts, CFRI records, and a questionnaire completed by parents of children with CF. All three of these sources are based on actual utilisation, which is a strength of the study. However, the parental questionnaire includes measures of utilisation that may suffer from recall bias since some of the measures are collected retrospectively. Additionally, as described in the methods section, where complete utilisation data were not available, assumptions were made. For example in assigning an average length of time to the time an individual child was on antibiotics. While the same assumptions were applied to both cohorts, these estimates could be considered a limitation of the study (Table 1).

Secondly, our study does not conduct a cost-benefit analysis, rather it restricts its focus to analysing cost differences between NBS and clinically diagnosed children. Thirdly, routine bronchoscopy was implemented in some Irish specialist CF centres in 2010/2011 during the sample period. Routine bronchoscopies should lead to an increased cost in the screened children who are diagnosed earlier, however, it was not implemented in all centres and may not have been implemented at the time of birth of all the screened children. Fourth, the study restricts its focus to Ireland. An issue to consider in comparing with results from other countries is the high prevalence of homozygous ΔF508 in the CF population. This is the most frequently occurring allele in most populations but is found at different frequencies depending on the ethnic group. In the USA, the frequency of the ΔF508 allele is approximately 70 % [26]. In contrast, a recent review of the CFRI data shows that the ΔF508 allele occurs with a frequency of 91.7 % (54.4 % homozygous) [27].

7. Conclusion

The results of this study imply that the benefits of newborn screening

Table 1
Summary of resource use, how costed and costs applied.

Resource Use	How costed	Cost estimation
Costs that were incurred both pre and post diagnosis - Parental questionnaire and CFRI/Hospital records		
Inpatient hospital admissions	HSE data	€825 per inpatient day
Public OPD Appointments	HSE data	€130 per OPD visit
Acute courses of oral antibiotics	Irish Medicines Formulary	Dependent on antibiotic received
Costs that were incurred post diagnosis - CFRI / Hospital records		
Day Case hospital admissions	HSE data	€735 with bronchoscopy, €637 without bronchoscopy
Acute Intravenous antibiotics	Hospital Pharmacy	Dependent on antibiotic received
Inhaled antibiotics	Irish Medicines Formulary	Cost of 28 days nebulised Tobramycin for positive pseudomonas aeruginosa cultures
Maintenance medications	Irish Medicines Formulary	Included pancreatic enzyme replacement and flucloxacillin
Costs that were incurred pre-diagnosis - Primarily parental questionnaire, supplemented by CRFI/hospital records as necessary		
Pre-diagnosis ED visits	HSE data	€268 per ED visit

CFRI = Cystic Fibrosis Registry of Ireland, HSE = Health Service Executive, OPD = Outpatient Department, ED = Emergency Department

Table 2
Descriptive characteristics of the newborn screening and clinical cohorts.

	NBS cohort (n=67)	Clinical cohort (n=72)	P value
% males	49.3	51.4	0.801
% ΔF508 homozygous	53.7	59.7	0.476
% diagnosed before 2 years of age	98.5	70.8	**<0.001**
Median (IQR) age in months at diagnosis	0.72 (0.66, 0.79)	4.44 (2.62, 13.72)	**<0.001**
Median (IQR) age in months between diagnosis and first encounter with CFRI	0 (0, 0.25)	0.48 (0, 0.77)	0.124
Median (IQR) age in months at last CFRI encounter before the age of 2	22.68 (22.00, 23.43)	22.92 (21.70, 23.47)	0.662

NBS: newborn screening cohort; CFRI: Cystic Fibrosis Registry of Ireland

for CF extend to the direct healthcare costs borne by the Irish State. The findings from this study can be partly transferable to other countries and will facilitate national and international policy makers to have a better understanding of the benefits of the CF NBS programme. Although this analysis took place in the pre-CFTR modulator era, this should not be considered a weakness. On the contrary, newborn screening may have even greater reduction in direct health care costs in the CFTR modulator era as children can be started on disease modifying therapies early and significantly reduce progression of bronchiectasis, a major driver of disease burden.

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Table 4
Comparison of mean direct medical cost per category between NBS and Clinically Diagnosed CF Cohorts.

	NBS cohort Mean (SD) *Median [IQR]*	Clinical cohort Mean (SD) *Median [IQR]*
Inpatient admissions	€5070 (€11,442) *€0* *[€0, €4170]*	€10,940 (€16,393) *€4080* *[€0, €14,928]*
Daycases	€254 (€501) *€0* *[€0, €648]*	€83 (€302) *€0* *[€0, €0]*
Pre-diagnosis emergency department visits	€17 (€65) *€0* *[€0, €0]*	€245 (€596) *€0* *[€0, €262]*
Public outpatient appointments	€1548 (€544) *€1452* *[€1190, €1844]*	€848 (€594) *€895* *[€31, €1269]*
Acute oral antibiotics	€47 (€64) *€24* *[€13, €65]*	€39 (€83) *€24* *[€0, €46]*
IV antibiotics	€60 (€172) *€0* *[€0, €15]*	€117 (€248) *€0* *[€0, €168]*
Inhaled antibiotics	€1093 (€2848) *€0* *[€0, €0]*	€993 (€2626) *€0* *[€0, €0]*
Maintenance medications	€1606 (€524) *€1823* *[€1639, €1854]*	€872 (€729) *€1059* *[€0, €1502]*
Total	€9598 (€13,928) *€4568* *[€3187, €11,021]*	€13,928 (€18,222) *€7876* *[€1139, €19,507]*

NBS = newborn screening; SD = standard deviation

Table 3
Costs per group by category.

	NBS cohort			Clinical cohort			p-value
	€0 N (%)	€1 - €1000 N (%)	>€1000 N (%)	€0 N (%)	€1 - €1000 N (%)	>€1000 N (%)	
Inpatient admissions*	40 (60%)	3 (4%)	24 (36%)	31 (43%)	1 (1%)	40 (56%)	**0.044**
Day cases**	50 (75%)	12 (18%)	5 (8%)	65 (90%)	6 (8%)	1 (1%)	**0.046**
Pre-diagnosis emergency department visits	61 (94%)	4 (6%)	-	40 (67%)	16 (27%)	4 (3%)	**<0.001**
Public outpatient appointments*	1 (2%)	5 (8%)	58 (91%)	17 (25%)	21 (31%)	30 (44%)	**<0.001**
Acute oral antibiotics*	3 (5%)	62 (95%)	-	16 (26%)	46 (74%)	-	**0.001**
IV antibiotics**	48 (72%)	19 (28%)	-	41 (57%)	29 (40%)	2 (3%)	0.090
Inhaled antibiotics**	53 (79%)	-	14 (21%)	60 (83%)	-	12 (16%)	0.828
Maintenance medications**	3 (4%)	4 (6%)	60 (90%)	22 (33%)	10 (15%)	34 (52%)	**<0.001**

* Includes pre and post diagnosis costs.
** Includes post diagnosis costs. NBS: newborn screening cohort.

Table 5
Unadjusted and adjusted regression analysis of total direct costs.

Unadjusted model (n=111)			
Predictor	**IRR**	**95% CI**	**p-value**
Clinical diagnosis	3.73	2.32, 5.97	<0.0001
NBS detection	ref		
Adjusted model (n=105)			
Predictor	**IRR**	**95% CI**	**p-value**
Clinical diagnosis	2.62	1.66, 4.15	<0.0001
NBS diagnosis	ref		
Homozygous ΔF508 or G511D	1.57	0.99, 2.50	0.056
No known homozygous mutation	ref		
Parental smoking	1.30	0.80, 2.11	0.290
Neither parent smokes	ref		
Maternal 3rd level degree	0.81	0.53, 1.22	0.309
Maternal education below 3rd level degree	ref		
Male sex	1.05	0.70, 1.59	0.798
Female sex	ref		
Duration between diagnosis and CFRI interaction (years)	0.54	0.33, 0.88	0.014

IRR = incidence rate ratio; NBS = newborn screening; CFRI = Cystic Fibrosis Registry of Ireland; ref = reference category

Conflict of interest statement

The authors have no conflict of interest to report.

Previous presentations of results

1. Somerville R, Fitzgerald C, George S, Linnane B, Segurado R, Kapur K, O'Ceilleachair A, Staines A, Fitzpatrick P. Comparison of direct healthcare costs in the first 2 years of life between screened and clinically diagnosed children with cystic fibrosis: the ICOS study. European Cystic Fibrosis Conference, Liverpool, June 2019. Journal of Cystic Fibrosis 2019; 18S1: 530
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CRediT authorship contribution statement

Rebecca Somerville: Investigation, Methodology, Writing – original draft, Writing – review & editing. **Catherine Fitzgerald:** Data curation, Project administration, Validation, Writing – review & editing. **Ricardo Segurado:** Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Kanika Kapur:** Formal analysis, Methodology, Writing – original draft, Writing – review & editing. **Sherly George:** Data curation, Project administration, Validation, Writing – review & editing. **Nancy Bhardwaj:** Writing – review & editing. **Barry Linnane:** Funding acquisition, Data curation, Writing – review & editing. **Alan O'Ceilleachair:** Methodology, Writing – review & editing. **Anthony Staines:** Writing – review & editing. **Patricia Fitzpatrick:** Conceptualization, Data curation, Formal analysis, Funding acquisition, Writing – original draft, Writing – review & editing.

Declaration of competing interest

We have no conflict of interest to disclose.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.jcf.2024.06.015.

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